B E N C H M A R K R E P O R T

The Benchmark That Broke Every Winning Streak

When Corrosion Resistance Blinds 13 Models to a Safety-Critical Trade-off

22	36.4%	0	5/13
LLMs Tested	Pass Rate	Models Pass All 5	Called FRP "Industry Standard"

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1. Executive Summary

I posed a walkway grating material selection question to 22 LLMs: which material is most appropriate for a personnel walkway between digesters, 15 meters above ground, in an H_2S and CH_4 environment? The correct answer is **C) Stainless Steel 304 or 316.**

The pass rate is 36.4% (8/22) — the second lowest in the series. But the real story is what this benchmark did to every pattern we had established across the previous four reports.

F1 — The "B" Trap — Corrosion Resistance vs. Safety: 13 models chose B (FRP/GRP), lured by its corrosion resistance, lightweight properties, and non-sparking behavior. But FRP has a critical weakness at 15m height: it is brittle — it breaks without warning, unlike steel which bends and gives visible deformation before failure. When personnel safety at height is involved, ductile failure mode matters more than corrosion resistance.

F2 — Every Winning Streak Broke: The two Qwen models (235B and Next 80B Instruct) that had passed all 4 previous benchmarks both failed here. Claude's 100% family consistency across 3 benchmarks collapsed to 1/5. DeepSeek V3.1, which had recovered from V3.2's failure, also lost one. No model has now passed all 5 WWTP benchmarks. Maximum score is 4/5, achieved by 5 different models — each with a different failure point.

F3 — Reversed Size Pattern: In previous benchmarks, larger/newer models generally outperformed smaller/older ones. Here, the pattern inverted: Gemini 2.0 Flash passed while Gemini 2.5 Pro and 3 Pro Preview failed. Gemma 4B passed while Gemma 12B failed. The models with less "sophistication" may have relied on simpler, more direct reasoning that happened to be correct.

F4 — Confidently Wrong: 5 of 13 models that chose B explicitly called FRP the "industry standard" for this application. This is not merely a wrong answer — it is a wrong answer delivered with authority. For industrial advisory contexts, confident incorrect recommendations may be more dangerous than uncertain ones.

F5 — Nobody Survives 5 Questions: Across the full WWTP series, 0 of 22 models have passed all 5 benchmarks. The funnel has closed completely: 22 → 17 → 15 → 2 → 0. Every model has at least one failure point in the series.

2. The Question

This benchmark tests whether models can weigh competing engineering criteria correctly. The scenario presents a walkway at 15 meters height between two digesters, with H_2S and CH_4 gases present, outdoor exposure, and regular personnel traffic. The correct answer requires understanding that corrosion resistance alone is not sufficient when human safety at height is involved.

Table 1: Answer Options and the Safety Trade-off

	Material	Strength	Assessment
A	Galvanized steel	Cheap, structural	WRONG — H_2S corrodes zinc coating rapidly
B	FRP/GRP	Corrosion-resistant, lightweight, non-sparking	**THE TRAP — brittle failure at height = unacceptable safety risk**
C	**Stainless Steel 304/316**	Corrosion-resistant AND ductile	**CORRECT — bends before breaking, safe failure mode**
D	PVC	No structural strength	WRONG — cannot support personnel loads

Table 1: Option B is the deliberate trap. FRP wins on corrosion resistance, weight, and spark safety. But it loses on the most critical criterion: failure mode at height. FRP shatters without warning; steel bends and deforms visibly before failure.

The key engineering insight: corrosion resistance is necessary but not sufficient. A plant operator who has seen FRP gratings crack under point loads (dropped tools, equipment wheels, concentrated foot traffic) would instinctively choose steel for a safety-critical elevated walkway. This is field knowledge that may not appear prominently in material specification databases.

3. Results

Figure 1: Answer Distribution

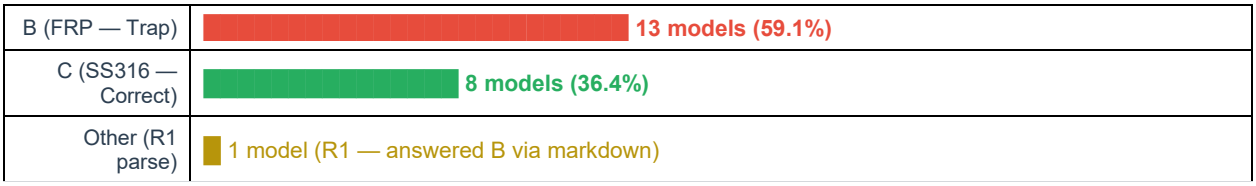

Figure 1: The majority (59%) fell into the FRP trap. No model chose A (galvanized) or D (PVC). The competition was entirely between B and C.

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Sonnet 4	C	PASS	348	8.8s	Claude
DeepSeek V3.1	C	PASS	408	3.8s	DeepSeek
DeepSeek V3.2	C	PASS	220	6.2s	DeepSeek

Gemini 2.0 Flash	C	PASS	559	5.0s	Gemini
Gemini 2.0 Flash Lite	C	PASS	341	3.4s	Gemini
Gemma 3 1B*	C	PASS	1	1.2s	Gemma
Gemma 3 4B	C	PASS	280	6.8s	Gemma
Gemma 3 27B	C	PASS	380	13.0s	Gemma
DeepSeek R1 (CoT)	**B†**	**FAIL**	947	6.5s	DeepSeek
Claude Haiku 4.5	**B**	**FAIL**	370	4.7s	Claude
Claude Opus 4.1	**B**	**FAIL**	372	12.2s	Claude
Claude Opus 4.5	**B**	**FAIL**	374	10.0s	Claude
Claude Sonnet 4.5	**B**	**FAIL**	356	9.9s	Claude
Gemini 2.5 Flash	**B**	**FAIL**	574	12.8s	Gemini
Gemini 2.5 Pro	**B**	**FAIL**	362	19.3s	Gemini
Gemini 3 Flash Preview	**B**	**FAIL**	443	9.7s	Gemini
Gemini 3 Pro Preview	**B**	**FAIL**	448	20.8s	Gemini
Gemma 3 12B	**B**	**FAIL**	210	12.3s	Gemma
Qwen3 235B A22B Instruct	**B**	**FAIL**	225	3.2s	Qwen
Qwen3 Coder 480B	**B**	**FAIL**	297	3.0s	Qwen
Qwen3 Next 80B Instruct	**B**	**FAIL**	227	4.4s	Qwen
Qwen3 Next 80B Thinking	**B**	**FAIL**	911	6.2s	Qwen

*Table 2: *Gemma 1B passed with single token, zero explanation. †R1 answered B via **B** markdown formatting (parsing issue + wrong answer).*

4. LLM Behavior Analysis

4.1 The "B" Trap — When Corrosion Resistance Blinds Models to Safety

Thirteen models chose FRP/GRP with remarkably consistent reasoning. The argument is technically sound on every point except the most critical one: failure mode at height.

Table 3: The Common Reasoning Chain in B-Choosing Models

Step	Reasoning	Assessment
1	H_2S + CH_4 = highly corrosive environment	Correct
2	FRP is corrosion-resistant, lightweight, non-sparking	Correct — these are real FRP advantages
3	Stainless steel is "expensive", "heavy", "susceptible to pitting"	**THE DISMISSAL — valid concerns but not decisive**
4	*FRP is "industry standard" for WWTP walkways*	**Stated by 5/13 models — confidently incorrect**
5	Conclusion: FRP is the best choice	Misses: brittle failure at 15m = unacceptable risk

The critical blind spot is failure mode. 8 of 13 B-choosers actually mentioned that FRP can be "brittle." But they treated it as a minor limitation rather than a disqualifying factor. In field

engineering practice, a material that shatters without warning at 15 meters height is not acceptable for a personnel walkway, regardless of its corrosion advantages. This is the kind of trade-off that operational experience teaches but material specification sheets may not emphasize.

> *FRP/GRP is the industry standard for digester and headworks applications in wastewater treatment plants specifically because of these corrosive gas environments.*
> — Claude Opus 4.5 — a confident but incorrect claim

A consistent trap pattern across the WWTP series: in Biogas, models followed “SOB = aerobic”; in Pump RCA, “high current = electrical”; in Dewatering, “vibration = bearings”; here, “corrosion = FRP.” Each trap exploits a shortcut where the model anchors on the most obvious requirement and fails to weigh competing criteria. The difference here is that the overlooked criterion is human safety, which arguably makes this the most consequential trap in the series.

4.2 Every Winning Streak Broke

This benchmark disrupted every consistency pattern from the previous four reports.

Table 4: Streak Disruptions

Pattern (Before)	Status Before Walkway	After Walkway
Qwen3 235B — perfect record	4/4 passed (only model to do so)	**FAILED — chose B**
Qwen3 80B Inst — perfect record	4/4 passed (only model to do so)	**FAILED — chose B**
Claude family — 100% (excl. Biogas)	5/5 on Pump RCA, Dewatering	**1/5 — only Sonnet 4 passed**
Qwen family — 100% (excl. Biogas)	4/4 on Pump RCA, Dewatering	**0/4 — complete family failure**
Larger = better	Generally held across benchmarks	**Inverted: Flash > Pro, 4B > 12B**

Table 4: This is why the guidelines say “observations, not conclusions.” Every pattern we identified proved to be question-dependent.

This is the strongest argument for the hedged language used throughout this series. In the Dewatering report, we wrote that “only 2 models have passed all 4 benchmarks.” Had we said “these 2 models are the most reliable,” this report would have immediately disproven it. The lesson: domain performance is question-dependent, and any claim of consistency is one benchmark away from being invalidated.

4.3 The Reversed Size Pattern — When Older Models Outperform Newer Ones

Table 5: Model Generation vs. Performance

Model	Answer	Status	Observation
Gemini 2.0 Flash	C	PASS	Older model, simpler reasoning → correct

Gemini 2.0 Flash Lite	C	PASS	Smallest Gemini, most concise → correct
Gemini 2.5 Flash	B	FAIL	Newer model with more reasoning → wrong
Gemini 2.5 Pro	B	FAIL	Flagship model → wrong
Gemini 3 Pro Preview	B	FAIL	Latest generation → wrong

Table 5: Within the Gemini family, model generation inversely correlates with performance on this question. A similar pattern appears in Gemma (4B pass, 12B fail).

One possible explanation: newer, more capable models may have more extensive training data about FRP in WWTP applications (where FRP is indeed commonly used for certain components). This broader knowledge may have reinforced the FRP association, while older models with less domain-specific training defaulted to a simpler "steel for structural safety" heuristic that happened to be correct here. This is speculative, but the pattern is notable.

4.4 Family Behavioral Patterns

Figure 2: Family Pass Rates

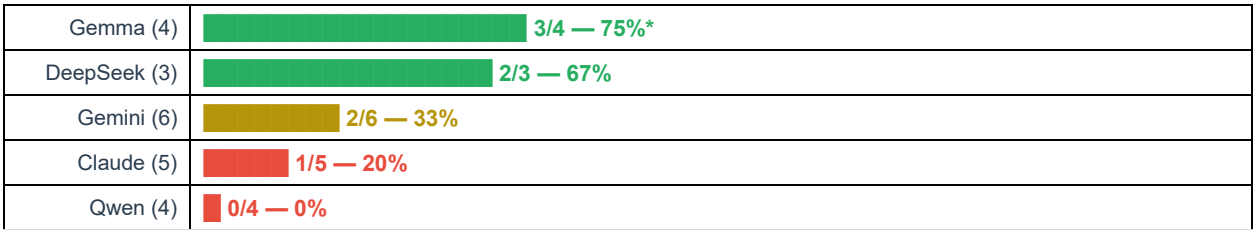

*Figure 2: Family rankings are completely inverted from previous benchmarks. Claude and Qwen, previously the strongest families, are now the weakest. *Gemma includes 1B single-token pass.*

The inversion is dramatic. Claude and Qwen had been 100% on Pump RCA and Dewatering. Here they are the worst performers. DeepSeek and Gemma, previously inconsistent, are the top families. This reinforces a growing observation: family-level performance is highly question-dependent and cannot be generalized from a small number of tests.

4.5 Confidently Wrong — The "Industry Standard" Claim

Five models explicitly stated that FRP is the "industry standard" for this application. This is not just a wrong answer — it is a wrong answer presented as established fact. In an industrial advisory context, a model that says "I'm not sure but maybe B" is less dangerous than one that says "B is the industry standard."

Table 6: Models That Claimed "Industry Standard"

Model	Claim
Claude Haiku 4.5	*"FRP is the industry standard for elevated walkways at WWTP facilities"*
Claude Opus 4.5	*"FRP is the industry standard for digester and headworks applications"*
Claude Sonnet 4.5	*"FRP/GRP is the industry standard for this exact application"*

Qwen3 Next 80B Instruct	*"making it the industry standard for elevated walkways in corrosive environments"*
Qwen3 Next 80B Thinking	*"industry standards explicitly recommend FRP/GRP for walkways in H_2S-rich areas"*

Table 6: These models did not hedge with "may be" or "is often used." They stated it as fact. This level of confidence in a wrong answer is a specific risk pattern worth tracking.

5. Cross-Benchmark Comparison — Five Questions, Zero Perfect Scores

Figure 3: The Narrowing Funnel — Now Fully Closed

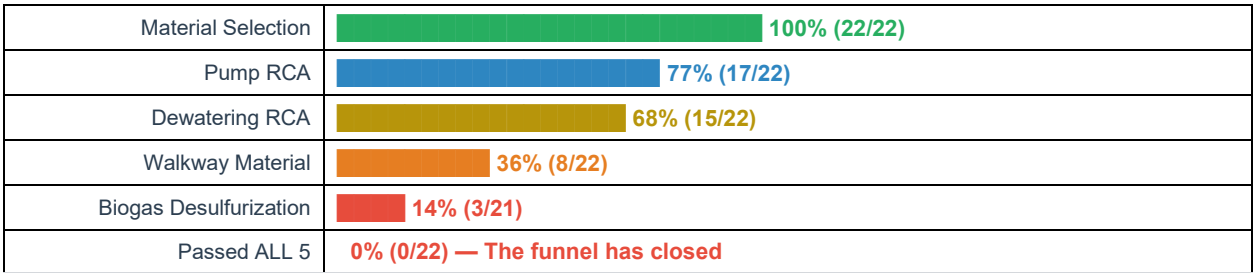

Figure 3: Five benchmarks, and no model has survived them all. The maximum score is 4/5, achieved by 5 different models — each with a different failure point.

Table 7: The 4/5 Club — Top Performers and Their Failure Points

Model	Material	Pump	Dewater	Walkway	Biogas
Qwen3 235B A22B Inst	✓	✓	✓	✗	✓
Qwen3 Next 80B Inst	✓	✓	✓	✗	✓
DeepSeek V3.2	✓	✓	✗	✓	✓
DeepSeek V3.1	✓	✓	✓	✓	✗
Claude Sonnet 4	✓	✓	✓	✓	✗

Table 7: Five models at 4/5, each failing on a different question. No two models share the same failure point. This suggests each question probes a different dimension of domain knowledge.

The pattern is telling: no two 4/5 models fail on the same question. Qwen models stumble on the safety/corrosion trade-off (Walkway). DeepSeek V3.2 stumbles on upstream reasoning (Dewatering). V3.1 and Claude Sonnet 4 stumble on deep domain knowledge (Biogas). Each benchmark probes a different cognitive dimension, and no model covers all of them.

Table 8: The WWTP Trap Taxonomy — Five Benchmarks, Five Reasoning Shortcuts

Benchmark	Trap Name	The Shortcut	What Was Missed
Biogas Desulfurization	**Aerobic Trap**	SOB = aerobic → aeration basin	Sulfur cycling occurs in digesters, not aeration
Pump RCA	**Electrical Trap**	High current + heat = winding damage	Mechanical overload also causes high current

Dewatering RCA	**Local Trap**	Vibration + scratches = bearing failure	Upstream grit causes the scratches that cause the vibration
Walkway Material	**Corrosion Trap**	H_2S environment = best corrosion resistance wins	Ductile failure mode matters more at 15m height
Material Selection	No trap	Binary choice, clear correct answer	(All models answered correctly)

Table 8: Each trap exploits a common pattern: the model anchors on the most salient feature (corrosion, current, vibration, aerobic conditions) and follows the most statistically likely association rather than reasoning from the full constraint set. The traps become more effective as the gap between the "obvious" answer and the correct answer widens.

6. Conclusions

This fifth benchmark may be the most important one in the series — not because of its pass rate, but because of what it did to every conclusion from the previous four reports.

The two models we highlighted as "series survivors" in the Dewatering report both failed here. The Claude family's consistency collapsed. The size-performance relationship inverted. These are not random fluctuations — they reveal that domain performance is fundamentally question-dependent, and any pattern observed across N questions may not hold at N+1.

The "B trap" adds a new category to the series' trap taxonomy: a trade-off trap where both options have genuine strengths, and the correct answer requires weighing competing engineering criteria rather than identifying one clearly correct path. This is arguably closer to real engineering practice, where decisions rarely have a single dominant option.

Perhaps the most practically relevant finding is the "confidently wrong" pattern: 5 models stated that FRP is the "industry standard" for this application. In an industrial advisory context, a model that is wrong and uncertain invites human verification. A model that is wrong and confident may bypass it. Tracking confidence calibration alongside accuracy may be worth incorporating into future benchmark analyses.

Looking back across five reports, the series has undergone a natural evolution. The early benchmarks asked "how well do LLMs perform on domain questions?" The later ones increasingly ask "what types of reasoning errors do they make, and how dangerous are those errors?" This shift — from measuring accuracy to characterizing failure modes — may be the most valuable outcome of the series so far. Accuracy tells you whether to trust a model. Failure mode analysis tells you where not to trust it.

7. Methodology Notes

Non-determinism: LLMs produce different outputs on repeated runs. This test used a single run per model. Results should be interpreted as indicative rather than absolute. Future iterations may incorporate multi-run protocols (3–5 runs) to assess stability.

Single question: This report analyzes one question. The dramatic reversal of patterns from previous reports underscores the importance of multi-question evaluation.

API conditions: All models were accessed through the Kaggle Benchmarks platform API with default temperature settings.

Correct answer rationale: C (Stainless Steel 304/316) is correct based on field engineering practice for elevated personnel walkways in WWTP environments. While FRP is used in WWTP settings for certain non-structural or ground-level applications, the combination of 15m height, regular personnel traffic, and the requirement for ductile failure modes makes stainless steel the appropriate choice for this specific scenario.

Cross-benchmark data: The 5-benchmark matrix references results from independently published reports. The Biogas benchmark used 21 models; all others used 22.